Campus Santa Fe

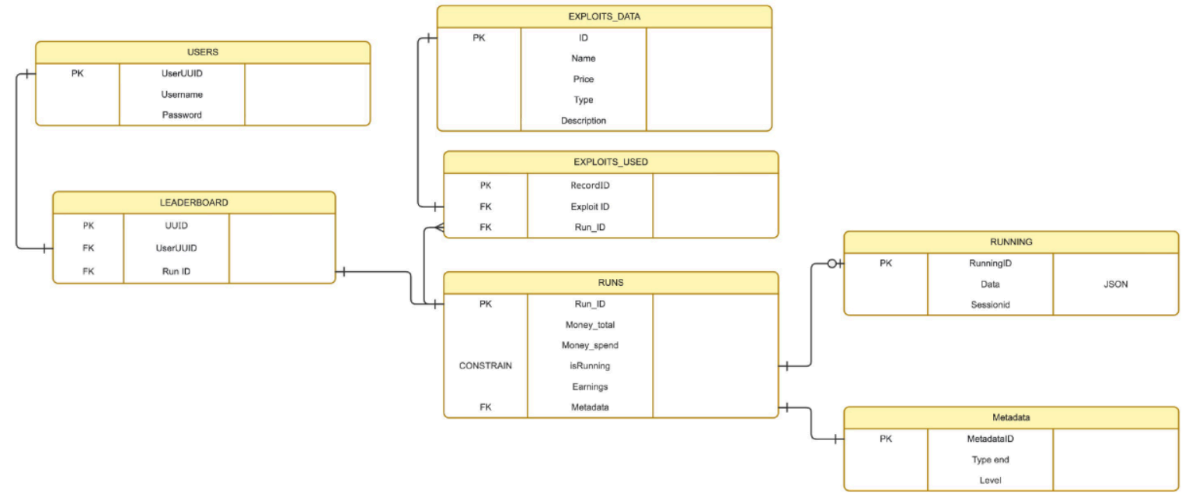
4. Database Normalization for the Project Supported by Generative AI Resources

Pablo Paz A01782158

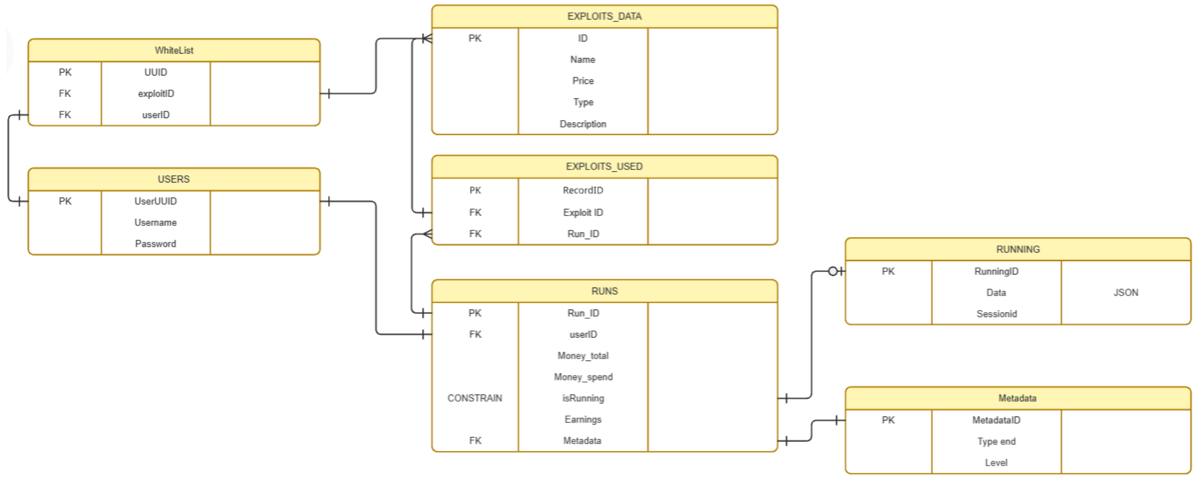
Nicolas Amaya A01787217

25 de mayo de 2026

Old db schema:



New db schema:



The database we designed was made following normalization principles up to the Third Normal Form (3NF) in order to reduce data redundancy, improve data integrity, and optimize overall system performance.

Each table represents a specific entity within the system. For example, `USERS` stores only user-related information, `EXPLOITS_DATA` contains the available exploit data, and `RUNS` stores information about game sessions. This separation ensures clear responsibilities for each entity and prevents unnecessary duplication of information.

The many-to-many relationship between runs and exploits was properly normalized through the `EXPLOITS_USED` junction table, avoiding the need to store multiple exploit values in a single column or repeat exploit information across records.

The `WhiteList` table acts as a relationship between users and the unlocked exploits while maintaining referential integrity through foreign keys.

The `Metadata` table remains as an independent entity because it stores serialized game states and snapshots used to speed up loading and recovery of saved runs. Although it may initially appear to contain derived data, this was made into another table as it was non-essential data related to the runs.

Likewise, the `RUNNING` table was designed as a fast-access layer for active game states. Its purpose is to allow quick recovery of a player's progress in case of disconnection, unexpected shutdown, or session interruption, effectively functioning as a persistent cache structure without violating normalization principles.

Thanks to this separation of entities, the model avoids incorrect transitive dependencies, eliminates unnecessary redundancy, and maintains consistency between tables through the use of primary keys and foreign keys. For these reasons, the database design can be considered properly normalized in Third Normal Form (3NF) while also maintaining an efficient and scalable architecture for handling runs and game state management.